

Ubuntu Security Review Lab - 9/15/26

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This lab was conducted in an Ubuntu virtual machine intentionally configured with several security misconfigurations using setup commands generated with assistance from ChatGPT. The objective was to analyze and interpret the configuration changes, investigate their effects within the Ubuntu environment, identify the resulting security concerns, and determine appropriate remediation and verification steps using Linux commands.

Before making configuration changes, a snapshot of the original VM was created to establish a recovery point in case the lab caused unintended system issues.

Lab Setup/Intentional Misconfigurations

Setup commands provided by ChatGPT:

- `sudo mkdir -p /opt/company`
- `sudo touch /opt/company/employee_records.txt`
- `sudo chmod 777 /opt/company/employee_records.txt`

- `sudo apt install apache2 -y`
- `sudo systemctl enable apache2`
- `sudo systemctl start apache2`

- `sudo mkdir -p /etc/company`
- `echo "backup_password=<LAB_PASSWORD>" | sudo tee /etc/company/backup.conf > /dev/null`
- `sudo chmod 644 /etc/company/backup.conf`

Scenario:

An Ubuntu workstation has been intentionally configured with potentially insecure settings. You've been asked to perform a basic security review using the provided configuration commands as a starting point. You know the system contains misconfigurations, but you have **not been told where they are**. Analyze the system and misconfiguration commands to determine what changes were made, assess whether those changes introduce security concerns, and remediate and verify each identified issue.

Investigation & Findings:

Finding 1

Finding: Unnecessary web service (Apache2) is active and configured to start automatically, VM is not intended for hosting web services, so Apache2 is unnecessary.

Security Impact: Running unnecessary services and applications increases attack surface by exposing additional software that can contain vulnerabilities.

Remediation: `sudo systemctl stop apache2 ; sudo systemctl disable apache2`

Verification: Used `systemctl status apache2` to verify the service was stopped and disabled.

Finding 2

Finding: `/opt/company/employee_records.txt` is a potentially sensitive file configured with full read, write, and execute permissions for the owner, group, and others (`777 / rwxrwxrwx`)

Security Impact: Allowing full permissions to all users risks compromising the confidentiality and integrity of the data, and violates least privilege.

Remediation: Update permissions. There is no need to execute this file, so `rw-r---` is appropriate: `sudo chmod 640 /opt/company/employee_records.txt`.

Verification: Confirmed permission change using `ls -l /opt/company/employee_records.txt`, which displayed `rw-r---` (640)

Finding 3

Finding: `/etc/company/backup.conf` contains a backup password and presents two security concerns.

(A) The file is configured with 644 permissions (`rw-r--r--`), allowing group and other users to read its contents.

(B) The second vulnerability is that the backup password itself is stored in plaintext within `backup.conf`.

Security Impact: Allowing read permissions for backup.conf for group and others potentially compromises the confidentiality of the password, and storing the password in plaintext within backup.conf is not a secure method of password storage.

Remediation: (A) Change the file permissions to 600 (rw-----) to restrict access to the owner;
sudo chmod 600 /etc/company/backup.conf

(B) Remove the backup password from the configuration file altogether and store it using a more secure mechanism.

Verification: Confirmed permission change using ls -l /etc/company/backup.conf to confirm the file permissions were changed to rw----- (600)

Conclusion:

This lab provided hands on experience analyzing Linux configuration changes, identifying security concerns, applying remediation, and verifying the results. The identified issues included an unnecessary running service, overly permissive file permissions, and insecure credential storage. Each finding was investigated and remediated using Linux command line tools and the principle of least privilege.

Skills Demonstrated:

- Linux filesystem navigation and file inspection
- Linux permissions and octal notation (777, 640, 644, 600)
- File permission management with chmod
- Service Investigation and management with systemctl
- Security concepts including least privilege, attack surface, confidentiality, and integrity
- Security finding documentation, remediation, and verification.